General Relativity: General Relativity Lecture 5

Leonard Susskind

Lecture notes organized by LazyingArt LLC with Video2Book

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Chapter 1

Light Cones, Geodesics, and the Schwarzschild Horizon

The horizon problem begins with a distinction already present in special relativity. Two nearby events may be separated in a time-like, space-like, or light-like direction. Curvature does not abolish this classification. It makes the light cone a local object, different at every event, and makes the metric itself dynamical. Once the metric is known, the motion of a freely falling particle follows from the same proper-time principle that describes a free relativistic particle in flat spacetime.

We shall use that principle twice. First it must reproduce Newtonian motion in a weak field. Then it will be applied to the Schwarzschild metric, where the meaning of time, radial distance, and velocity becomes unexpectedly subtle.

1.1 Time-like, Space-like, and Light-like Intervals

In flat spacetime, with $c = 1$, the proper-time element is

$$d\tau^2 = dt^2 - dx^2 - dy^2 - dz^2. \tag{1.1}$$

Restoring c is useful whenever a nonrelativistic approximation is near:

$$d\tau^2 = dt^2 - \frac{dx^2 + dy^2 + dz^2}{c^2}. \tag{1.2}$$

The notation $d\tau^2$ does not guarantee that the right-hand side is positive. Its sign classifies the displacement:

$$d\tau^2 > 0 \quad \Longleftrightarrow \quad \text{time-like,} \tag{1.3}$$
$$d\tau^2 = 0 \quad \Longleftrightarrow \quad \text{light-like,} \tag{1.4}$$
$$d\tau^2 < 0 \quad \Longleftrightarrow \quad \text{space-like.} \tag{1.5}$$

For a space-like displacement it is convenient to define the positive proper distance

$$ds^2 = -d\tau^2 = dx^2 + dy^2 + dz^2 - dt^2 \qquad (c = 1). \tag{1.6}$$

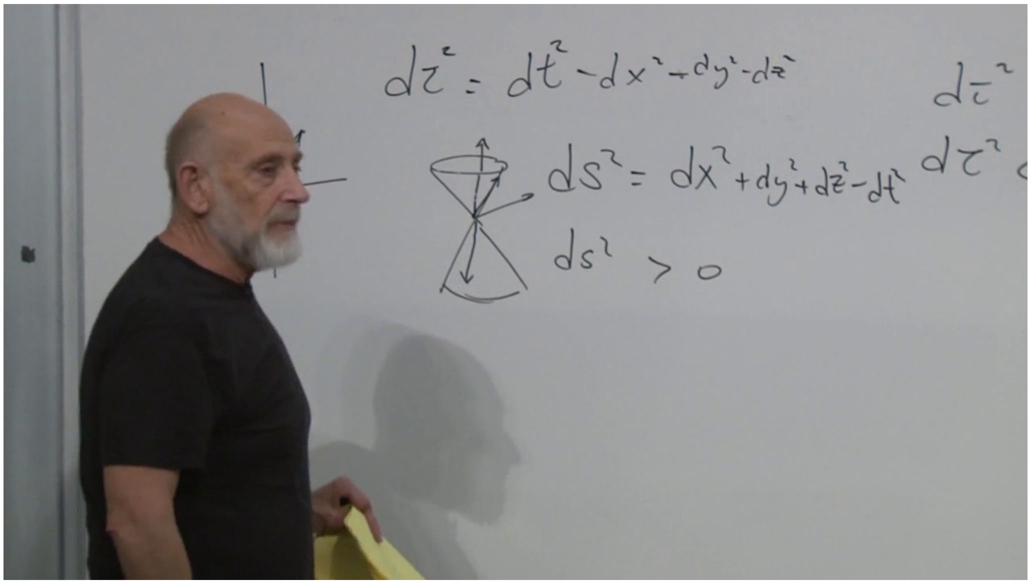

Figure 1.1: Proper time, proper distance, and the causal cone assembled on the board. Light rays occupy the cone itself; the interior is time-like and the exterior is space-like.

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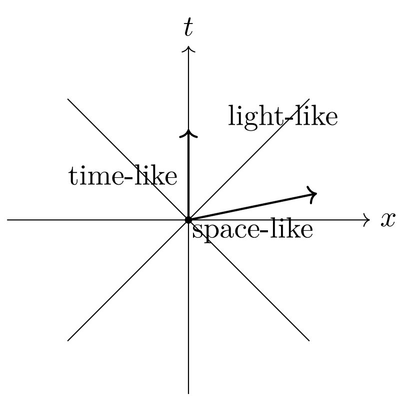

Figure 1.2: The causal classification in a two-dimensional spacetime section.

The light cone organizes these three possibilities. Its surface, not its filled interior, consists of the light-like directions. Time-like directions lie inside the cone and space-like directions lie outside. The upper and lower halves represent the future and past light cones.

These intervals are operational. Proper time is what a clock carried along a time-like worldline reads. Proper distance is what a suitable set of meter sticks measures between space-like separated events. They are not merely names for signs in a quadratic form.

1.2 Lorentzian Signature and a Cone at Every Event

Using the mostly-plus tensor convention, the flat metric has components

$$\eta_{\mu\nu} = \mathrm{diag}(-1, 1, 1, 1), \tag{1.7}$$

and

$$d\tau^2 = -\eta_{\mu\nu}\, dx^\mu dx^\nu. \tag{1.8}$$

The displayed matrix is the expression of a rank-two tensor in one especially simple basis. Its invariant feature is its signature: one negative and three positive directions,

$$\mathrm{signature}(\eta) = (-, +, +, +). \tag{1.9}$$

A metric with two negative signs would have two temporal directions. Such signatures can be studied mathematically, but they do not describe the spacetime assumed here.

In curved spacetime the components vary:

$$g_{\mu\nu} = g_{\mu\nu}(x). \tag{1.10}$$

At each event the metric must still have Lorentzian signature. Consequently, each event has its own time-like directions, light-like boundary, and space-like directions. A coordinate drawing may show cones tilted or squashed from point to point, but the causal classification remains.

Question from the audience. Does the single negative eigenvalue select one unique time direction?

Response. A basis that diagonalizes the displayed matrix has one negative basis direction, but that does not select a preferred observer or a unique physical arrow. Lorentz transformations mix the time coordinate with space coordinates, and every vector inside the cone is time-like. The invariant object is the cone together with the one-minus, three-plus signature, not one distinguished time-like vector. [a]

[a] Lecture timestamp: 00:12:17–00:17:24.

Question from the audience. Why can two drawings of the local cone have different shapes?

Response. Coordinates can change the drawing. For example, replacing x by a rescaled coordinate changes the numerical slope assigned to a light ray and therefore makes the cone appear narrower or wider. More general coordinates can tilt it. The same event and the same causal relations are being represented. In curved spacetime cones at distinct events also belong to distinct tangent spaces, so their comparison requires the metric and a transport rule. [a]

[a] Lecture timestamp: 00:13:37–00:15:59.

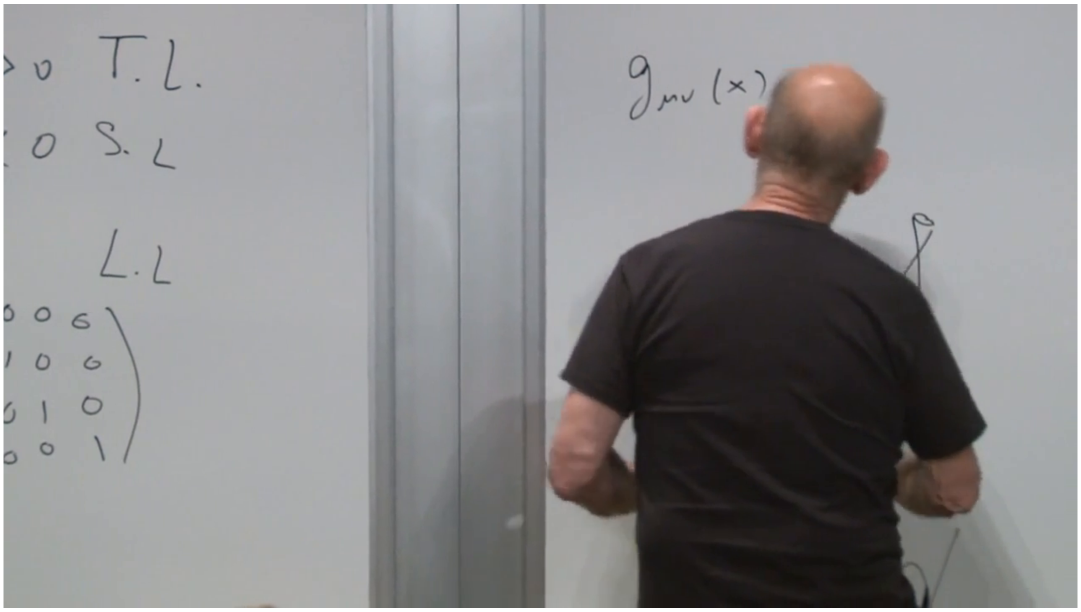

Figure 1.3: The position-dependent metric $g_{\mu\nu}(x)$, the causal labels, and sketches of local light cones.

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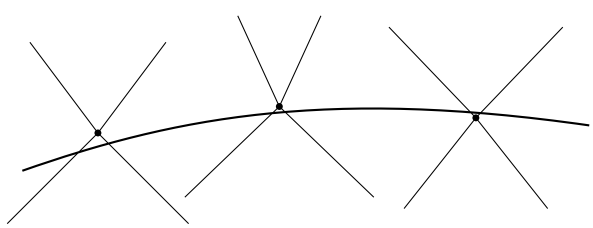

Figure 1.4: Neighboring local cones. Their coordinate shape can change without changing which vectors are time-like, light-like, or space-like.

1.3 Geodesics as a Variational Problem

A freely moving particle follows a spacetime geodesic. In differential form, using an affine parameter λ,

$$\frac{d^2x^\mu}{d\lambda^2} + \Gamma^\mu{}_{\rho\sigma}\frac{dx^\rho}{d\lambda}\frac{dx^\sigma}{d\lambda} = 0. \tag{1.11}$$

The connection tells the tangent how to remain parallel to itself as the particle moves.

There is a second formulation that is often easier to calculate with. In a Riemannian space with metric g_{mn}, the length of a curve is

$$s = \int ds = \int \sqrt{g_{mn}\, dx^m dx^n}. \tag{1.12}$$

The geodesic makes this length stationary under small variations of the path with fixed endpoints. The integral is simply the sum of infinitesimal lengths along the curve, and its Euler–Lagrange equation is a second-order differential equation analogous in form to a force law.

For a time-like worldline, spatial length is replaced by proper time:

$$\tau = \int \sqrt{-g_{\mu\nu}\, dx^\mu dx^\nu}. \tag{1.13}$$

A time-like geodesic makes τ stationary and, for nearby freely falling paths before conjugate points occur, locally maximal. Multiplying by $-m$ gives the particle action

$$A = -m\tau = -m\int \sqrt{-g_{\mu\nu}\, dx^\mu dx^\nu}. \tag{1.14}$$

Thus the same path locally minimizes $A = -m\tau$. The distinction between a stationary proper time and the sign of the chosen action removes the apparent minimum-versus-maximum ambiguity.

1.4 From Proper Time to an Ordinary Lagrangian

The action in (1.14) is geometric, but it can be put in the familiar form used in mechanics. Parameterize the worldline by the coordinate t, divide each differential by dt, and pull the remaining factor of dt outside the square root:

$$A = -m\int \sqrt{-g_{\mu\nu}(x)\, dx^\mu dx^\nu} \tag{1.15}$$

$$= -m\int \sqrt{-g_{\mu\nu}(x)\frac{dx^\mu}{dt}\frac{dx^\nu}{dt}}\, dt. \tag{1.16}$$

For the temporal component $dt/dt = 1$; the spatial components are ordinary coordinate velocities. We may therefore write

$$A = \int L\, dt, \qquad L = -m\sqrt{-g_{\mu\nu}(x)\dot{x}^\mu\dot{x}^\nu}, \qquad \dot{x}^\mu = \frac{dx^\mu}{dt}. \tag{1.17}$$

Question from the audience. Should the spacetime indices in this action be Greek?

Response. Yes. Greek indices run over time and all spatial coordinates. Latin indices may be reserved for the spatial coordinates when that distinction is useful. [a]

[a] Lecture timestamp: 00:25:53–00:26:03.

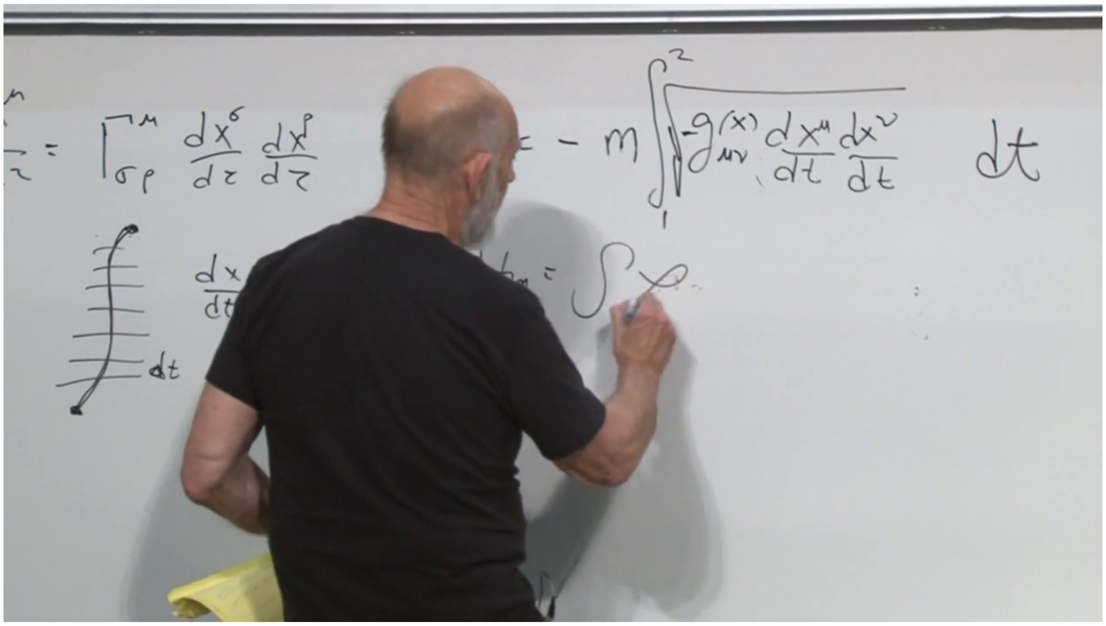

Figure 1.5: The worldline action being converted into the conventional form $A = \int L\,dt$.

Lecture frame: 00:27:00.

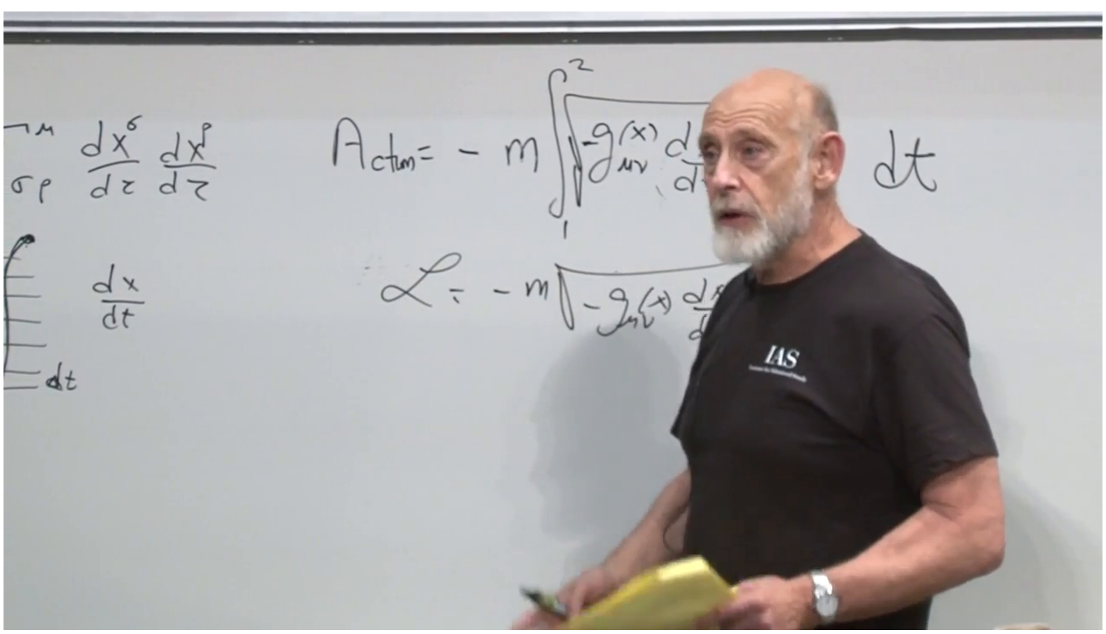

Figure 1.6: A later board state with the relativistic Lagrangian isolated.

Lecture frame: 00:28:59.

Question from the audience. Why is the square-root expression called the Lagrangian?

Response. After the action has been written as $A = \int L\,dt$, the integrand is the Lagrangian. This is not only a definition: varying that integrand gives the Euler–Lagrange equations

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{x}^\mu}\right) = \frac{\partial L}{\partial x^\mu},$$

which are equivalent to the geodesic equation after the parameterization is handled consistently. [a]

[a] Lecture timestamp: 00:28:36–00:29:02.

For a massive particle the worldline is time-like, so the quantity under the square root is positive. A space-like path would describe superluminal motion, not the trajectory of an ordinary massive particle. Given a metric, one can now obtain the particle's path either by computing the Christoffel symbols and using (1.11), or by applying Euler–Lagrange to (1.17). Showing their equivalence is a useful exercise; the action route is usually shorter.

1.5 The Newtonian Limit

Consider the weak, static gravitational field outside a spherical source. Begin with flat spacetime and allow the coefficient of dt^2 to differ slightly from one:

$$d\tau^2 = \left(1 + \frac{2u(\mathbf{x})}{c^2}\right) dt^2 - \frac{d\mathbf{x}^2}{c^2}, \tag{1.18}$$

where u is to become the Newtonian potential. This ansatz must be checked, not merely named.

The dimensionful particle action is

$$A = -mc^2 \int \sqrt{1 + \frac{2u}{c^2} - \frac{\dot{\mathbf{x}}^2}{c^2}}\, dt. \tag{1.19}$$

Both u/c^2 and $\dot{\mathbf{x}}^2/c^2$ are small. Keeping their first-order contributions,

$$\sqrt{1+\epsilon} = 1 + \frac{\epsilon}{2} + O(\epsilon^2), \qquad \epsilon = \frac{2u - \dot{\mathbf{x}}^2}{c^2}, \tag{1.20}$$

gives

$$L = -mc^2 \sqrt{1 + \frac{2u - \dot{\mathbf{x}}^2}{c^2}} \tag{1.21}$$

$$= -mc^2 - mu + \frac{m}{2}\dot{\mathbf{x}}^2 + O(c^{-2}u^2, c^{-2}u\dot{\mathbf{x}}^2, c^{-2}\dot{\mathbf{x}}^4). \tag{1.22}$$

The constant rest-energy term has no effect on the equations of motion. What remains is

$$L_{\mathrm{nr}} = \frac{m}{2}\dot{\mathbf{x}}^2 - mu(\mathbf{x}), \tag{1.23}$$

and therefore

$$m\ddot{\mathbf{x}} = -m\boldsymbol{\nabla} u. \tag{1.24}$$

The test mass cancels, as it must for universal free fall. The potential energy is mu, while the acceleration is independent of m.

We have consequently identified the leading weak-field relation

$$g_{00} = 1 + \frac{2u}{c^2} + O(c^{-4}). \tag{1.25}$$

Outside a spherical mass M,

$$u(r) = -\frac{GM}{r}, \qquad g_{00} = 1 - \frac{2GM}{rc^2} + O(c^{-4}). \tag{1.26}$$

This determines the first correction to the time component. It does not yet determine the complete strong-field metric.

1.6 Spherical Coordinates and the First Schwarzschild Guess

Spherical coordinates are natural for a central source. The lecture measures θ from the equatorial plane rather than from the pole, so its angular line element is

$$d\Omega^2 = d\theta^2 + \cos^2\theta\, d\phi^2. \tag{1.27}$$

With this convention,

$$d\mathbf{x}^2 = dr^2 + r^2 d\Omega^2. \tag{1.28}$$

Nothing physical depends on choosing this latitude-like θ; using the more common polar angle would replace $\cos^2\theta$ by $\sin^2\theta$.

Inserting the Newtonian time coefficient while leaving the spatial metric flat gives the first guess

$$d\tau^2 = \left(1 - \frac{r_s}{r}\right) dt^2 - \frac{1}{c^2}\left(dr^2 + r^2 d\Omega^2\right), \qquad r_s = \frac{2GM}{c^2}. \tag{1.29}$$

This cannot be the exact answer. For $r < r_s$, the coefficient of dt^2 changes sign while every spatial coefficient remains negative in the proper-time convention. The metric would lose its one-time, three-space signature.

The vacuum field equations supply the required radial coefficient:

$$\boxed{d\tau^2 = \left(1 - \frac{r_s}{r}\right) dt^2 - \frac{dr^2}{c^2\left(1 - r_s/r\right)} - \frac{r^2}{c^2} d\Omega^2.} \tag{1.30}$$

This is the exterior Schwarzschild metric. When r passes through r_s, the t and r coefficients change sign together, preserving Lorentzian signature. In Schwarzschild coordinates the radial coordinate becomes time-like inside and the coordinate t becomes space-like. This exchange signals a failure of the coordinate chart at the horizon, not a local failure of the spacetime geometry.

Question from the audience. How long does a falling object take to cross the horizon?

Response. Measured by the Schwarzschild coordinate t used by a distant observer, the crossing is pushed to infinite time. Measured by the proper time on the falling object's clock, it occurs in finite time. The distinction between these clocks is the central horizon puzzle. [a]

[a] Lecture timestamp: 00:54:45–00:55:35.

Question from the audience. Why alter the radial coefficient rather than some other part of the metric?

Response. The metric must retain Lorentzian signature, spherical symmetry distinguishes the radial direction from the angular sphere, and the vacuum field equations determine the reciprocal factor precisely. Signature alone motivates where to look; Einstein's equations give the exact result. [a]

[a]Lecture timestamp: 00:55:35–00:56:01.

For the Earth, r_s is only of order a centimeter, deep inside the material body. The exterior vacuum metric is not valid there, so the Earth has no Schwarzschild horizon. An interior solution must replace it inside the matter and match it at the surface.

Question from the audience. If t and r exchange coordinate roles, what happens to an ordinary clock?

Response. The clock continues to measure proper time along a time-like worldline. It does not know that a chosen coordinate has become awkward. A regular freely falling coordinate system shows smooth local geometry and ordinary local light propagation through the horizon. [a]

[a]Lecture timestamp: 00:56:35–00:57:50.

1.6.1 Why the radial correction did not appear in Newton's equation

The exact metric changes the radial coefficient already at first order in the potential:

$$\frac{1}{1-r_s/r} = 1+\frac{r_s}{r}+O\left(\frac{r_s^2}{r^2}\right) = 1-\frac{2u}{c^2}+O(c^{-4}). \tag{1.31}$$

Why was it legitimate to ignore that change in the Newtonian derivation? Inside the particle action it multiplies $\dot{r}^2/c^2$. Its first correction is therefore proportional to

$$\frac{u\,\dot{r}^2}{c^4}, \tag{1.32}$$

which is beyond the order retained in (1.22). The radial correction is negligible for the slow-motion Newtonian test, yet it becomes decisive for relativistic motion and near the horizon. This is a useful warning: a term too small for one approximation may control the phenomenon of interest in another.

1.7 Radial Fall in Schwarzschild Time

Set $c = 1$ and first study radial motion. Circular orbits are also important, but they require the angular terms and are postponed. Radial motion is less an orbit than a direct crash toward the source, and it exposes the horizon behavior with the least algebra.

Question from the audience. Whose time appears in the radial velocity?

Response. The dot below means differentiation with respect to the Schwarzschild coordinate time t, the time normalized to the distant static observer. The falling observer's wristwatch measures proper time τ. Confusing the two would turn a coordinate statement into a false local one. [a]

[a]Lecture timestamp: 01:04:03–01:04:47.

For $d\Omega = 0$, define

$$a(r) = 1 - \frac{r_s}{r}. \tag{1.33}$$

The radial metric and Lagrangian are

$$d\tau^2 = a\,dt^2 - \frac{dr^2}{a}, \tag{1.34}$$

$$L = -m\sqrt{a - \frac{\dot{r}^2}{a}}, \qquad \dot{r} = \frac{dr}{dt}. \tag{1.35}$$

Because L has no explicit t-dependence, its Hamiltonian is conserved. The radial momentum is

$$p_r = \frac{\partial L}{\partial \dot{r}} = \frac{m\dot{r}}{a\sqrt{a - \dot{r}^2/a}}, \tag{1.36}$$

and

$$H = p_r\dot{r} - L \tag{1.37}$$

$$= \frac{ma}{\sqrt{a - \dot{r}^2/a}} = E. \tag{1.38}$$

Solving for the coordinate velocity gives

$$\boxed{\dot{r}^2 = a^2\left(1 - \frac{m^2}{E^2}a\right).} \tag{1.39}$$

The board calculation tests several intermediate algebraic forms; the boxed expression is the consistent result.

As r approaches r_s, $a(r)$ approaches zero while the factor in parentheses approaches one. Hence

$$\frac{dr}{dt} \sim -\left(1 - \frac{r_s}{r}\right) \longrightarrow 0 \qquad (r \longrightarrow r_s^+) \tag{1.40}$$

for an ingoing particle. In Schwarzschild time, the particle appears to slow and freeze just outside the horizon.

Question from the audience. Does the decreasing value of dr/dt mean that gravity has become repulsive?

Response. No. The particle is not locally pushed outward. The coordinate t becomes increasingly ill-suited to describing the crossing, so radial progress per unit t tends to zero. Proper acceleration along the freely falling worldline remains zero, and the horizon is reached in finite proper time. [a]

[a] Lecture timestamp: 01:14:18–01:16:38.

Indeed, near the horizon

$$\frac{dt}{dr} \sim -\frac{1}{1 - r_s/r} = -\frac{r}{r - r_s}. \tag{1.41}$$

Its integral diverges logarithmically. This proves the infinite Schwarzschild-time statement without implying an infinite reading on the falling clock.

1.8 Radial Light Rays

Light gives the shortest version of the same calculation. For a radial null ray,

$$d\tau^2 = a\,dt^2 - \frac{dr^2}{a} = 0. \tag{1.42}$$

Therefore

$$\boxed{\frac{dr}{dt} = \pm a(r) = \pm\left(1 - \frac{r_s}{r}\right).} \tag{1.43}$$

The plus sign describes an outgoing ray and the minus sign an ingoing ray. Both coordinate speeds tend to zero at the horizon.

Question from the audience. What speed does the formula give far from the source?

Response. As r tends to infinity, $a(r)$ tends to one, so dr/dt tends to ± 1, or $\pm c$ when units are restored. [a]

[a]Lecture timestamp: 01:21:01–01:21:29.

General relativity is often like this: the principles fit in a few lines, while coordinates and calculations grow complicated. Equation (1.43) is simple, but interpreting its vanishing near r_s requires care. It is a coordinate speed, not the speed measured by a local inertial observer.

1.9 What a Distant Observer Sees

Question from the audience. Can light emitted near the horizon escape, and is that Hawking radiation?

Response. An outgoing classical ray emitted just outside the horizon can move outward, but its Schwarzschild coordinate speed begins extremely small and the signal reaches infinity only after a severe delay and redshift. A ray on the horizon remains on the horizon in these coordinates; a ray inside cannot escape. Hawking radiation is a quantum-field effect associated with the exterior geometry, not a classical light ray climbing out from inside, and it is not derived by the present calculation. [a]

[a]Lecture timestamp: 01:22:08–01:24:28.

Question from the audience. If a photon looks almost stationary, does it begin to look massive?

Response. No. A freely falling local observer always measures the photon moving at c. Only the coordinate ratio dr/dt tends to zero. Mass, nullness, and the local speed of light are not changed by a bad coordinate chart. [a]

[a]Lecture timestamp: 01:24:31–01:25:26.

The Schwarzschild expression used here is a vacuum exterior solution. A star, planet, or collapsing cloud has a different interior metric that must be matched to it. If the material surface remains outside r_s, as the Earth's does by an enormous margin, the exterior coordinate horizon lies outside the domain where the vacuum formula can be continued through the matter. If the body collapses inside its Schwarzschild radius, the horizon becomes part of the physical spacetime.

Question from the audience. If an outside observer never sees matter cross, how can a black hole ever form or grow?

Response. The premise uses Schwarzschild time as though it were a global physical clock at the horizon. The conclusion that a black hole can therefore never grow sounds plausible but is wrong. A complete answer requires regular horizon-crossing coordinates and the global causal geometry of collapse; that resolution is deferred to the next stage of the course. [a]

[a]Lecture timestamp: 01:27:57–01:29:36.

Question from the audience. What would a radar gun report for a car falling toward the horizon?

Response. The coordinate speed first increases as the car falls, then turns over and decreases toward zero. Radar pulses sent later can still catch the car outside the horizon, but the round trip takes progressively longer. The returned signal is increasingly redshifted, and the image appears slowed and compressed near r_s. This is not a reliable local picture of what the driver experiences; the driver crosses in finite proper time. [a]

[a]Lecture timestamp: 01:29:39–01:32:14.

Question from the audience. Is the speed of light constant only in special relativity?

Response. It is constant in every local inertial frame in general relativity as well. An observer falling with local clocks and meter sticks measures a nearby photon at c. Coordinate speeds such as dr/dt may vary because r and t are not the readings of one local inertial laboratory over the whole spacetime. [a]

[a]Lecture timestamp: 01:32:20–01:34:20.

1.10 Questions Beyond the Schwarzschild Calculation

Question from the audience. What might it mean to unify gravity with the other forces?

Response. One geometric possibility is to enlarge spacetime with additional compact directions. Components of a higher-dimensional metric can then appear in four dimensions as gauge fields and scalar fields, while momentum in an extra direction can appear as a charge. This is a structural hint associated with Kaluza–Klein ideas, not a completed unification in the present lecture. [a]

[a]Lecture timestamp: 01:34:22–01:36:08.

Question from the audience. Could different regions or universes have different metric signatures?

Response. Classical general relativity ordinarily fixes a nondegenerate Lorentzian signature. Changing it would require the metric to pass through a degenerate configuration, where the standard classical description fails. Quantum gravity may force a broader language, but there is no established theory here that predicts physical signature-changing regions. [a]

[a]Lecture timestamp: 01:36:08–01:38:08.

Question from the audience. How did Schwarzschild find this solution while serving during the First World War?

Response. The point made in class is human as much as historical: Schwarzschild was an astronomer and a highly accomplished mathematician, and working through Einstein's new equations offered an intellectual refuge amid wartime service. [a]

[a] Lecture timestamp: 01:38:11–01:39:02.

The central lesson is now sharp enough to carry forward. The metric determines both causal cones and free motion. Its weak-field time component contains Newton's potential, while its exact Schwarzschild form gives a coordinate horizon at $r = r_s$. Schwarzschild time stretches the crossing to infinity; the falling observer's proper time does not. Resolving that contrast globally, and explaining how a black hole forms despite the frozen distant image, is the next problem.